PHARMACOLOGY · MCQ PRACTICE

Female Sex Hormones & Contraception — MCQ Practice

SGL 1 · Dr. Chro Yaseen Qader
GUE BLOCK

Pharmacology GUE Block · MEDucation by Mattin Majid

HOW TO USE THIS SET
20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.
4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. Which ovarian structure secretes progesterone during the luteal (second) half of the menstrual cycle? EASY

A. Corpus albicans
B. Graafian follicle
C. Primordial follicle
D. Corpus luteum
E. Theca interna only

2. Which synthetic oestrogen is most commonly used in combined oral contraceptive pills? EASY

A. Mestranol
B. Oestriol
C. Conjugated equine oestrogens
D. Diethylstilbestrol
E. Ethinylestradiol

3. For maximum effectiveness, levonorgestrel emergency contraception should be taken as soon as possible and preferably within how many hours of unprotected intercourse? EASY

A. 72 hours
B. 24 hours
C. 96 hours
D. 48 hours
E. 120 hours

4. In a woman with an intact uterus, giving unopposed oestrogen (without a progestogen) as replacement therapy chiefly risks which complication? EASY

A. Ovarian failure
B. Endometrial hyperplasia
C. Osteoporosis
D. Hypertriglyceridemia only
E. Vaginal atrophy

5. A woman is prescribed a monophasic combined pill. Compared with a triphasic preparation, what distinguishes the monophasic regimen? MEDIUM

A. It contains progestin only, with no oestrogen component
B. It delivers a constant dose of oestrogen and progestin throughout the 21–24 active days
C. It is taken continuously with no placebo/hormone-free interval
D. It delivers increasing progestin doses over three 7-day periods to mimic the natural cycle
E. It is given as a single depot injection rather than daily tablets

6. A woman with acne and hirsutism wants a combined pill with an anti-androgenic progestin component. Which agent fits best? MEDIUM

A. Norethisterone
B. Norgestrel
C. Levonorgestrel
D. Medroxyprogesterone acetate
E. Drospirenone

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7. A woman on combined oral contraceptives starts rifampin for tuberculosis treatment and later presents pregnant despite reportedly good pill compliance. What best explains the contraceptive failure? HARD

A. Rifampin reduces gastrointestinal motility, preventing pill absorption
B. Rifampin suppresses ovarian oestrogen synthesis directly
C. Rifampin binds and inactivates circulating progestin
D. Rifampin induces hepatic enzymes that increase oestrogen/progestin metabolism
E. Rifampin competes with the pill for sex-hormone-binding globulin

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8. A postmenopausal woman with osteoporosis and a high risk of breast cancer is being started on chemoprevention. Why might raloxifene be preferred over tamoxifen in this specific patient? HARD

A. Raloxifene does not increase thromboembolism risk
B. Raloxifene has no effect on bone density
C. Raloxifene is anti-oestrogenic on the endometrium too, so it does not raise endometrial cancer risk while still protecting bone
D. Raloxifene is not a SERM and acts through a different receptor entirely
E. Raloxifene has no effect on breast tissue at all

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9. A breastfeeding mother who smokes wants contraception and cannot tolerate oestrogen. Which option is most appropriate? MEDIUM

A. Progestin-only pill
B. Transdermal combined patch
C. Combined monophasic pill
D. Combined triphasic pill
E. Combined vaginal ring

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10. Approximately how long does a single subdermal etonogestrel contraceptive implant provide contraception? MEDIUM

A. About 3 years
B. About 5 years
C. About 10 years
D. About 6 months
E. About 1 year

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11. Which patient history is a contraindication to placement of an intrauterine contraceptive device? MEDIUM

A. Mild obesity
B. History of migraine without aura
C. Family history of breast cancer
D. History of ectopic pregnancy
E. Smoking

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12. By what mechanism does the progestin component of combined contraceptives contribute to preventing pregnancy, independent of ovulation suppression? MEDIUM

A. Accelerating endometrial shedding mid-cycle
B. Directly lysing sperm on contact
C. Blocking fertilisation at the level of the zona pellucida
D. Thickening cervical mucus to impede sperm transport
E. Inducing tubal spasm to trap the oocyte

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13. Mifepristone is combined with which second agent for medical termination of pregnancy? MEDIUM

A. Letrozole
B. Medroxyprogesterone acetate
C. Clomiphene
D. Misoprostol
E. Levonorgestrel

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14. A woman presents 100 hours (just over 4 days) after unprotected intercourse requesting emergency contraception. Levonorgestrel is now outside its effective window. Which agent can still be offered? MEDIUM

A. Letrozole
B. Mifepristone
C. Clomiphene
D. Conjugated oestrogens
E. Ulipristal

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15. "Conjugated estrogens," used in hormone replacement therapy, are best described as: MEDIUM

A. A mixture of several natural estrogen hormones
B. A selective estrogen receptor modulator
C. An aromatase inhibitor
D. An anti-progestogen
E. A single synthetic ultra-potent estrogen

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16. A patient receiving injectable medroxyprogesterone acetate for contraception should be counselled that, after stopping the injections, she may experience: MEDIUM

A. Increased, not decreased, fertility initially
B. No effect on fertility timing at all
C. Permanent infertility
D. Immediate return of fertility within days
E. A delay of several months before fertility returns

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17. A woman with a personal history of oestrogen-receptor-positive breast cancer is requesting contraceptive advice. Combined hormonal contraceptives are contraindicated here primarily because: HARD

A. Progestin causes direct tumour lysis
B. Combined pills cause irreversible hepatotoxicity
C. They are only contraindicated in cervical, not breast, cancer
D. They sharply raise blood glucose in all patients
E. Oestrogen can stimulate growth of oestrogen-dependent neoplasms

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18. A 30-year-old with a history of migraine with aura asks about starting combined oral contraceptives. What is the main concern? HARD

A. Migraine has no bearing on contraceptive choice
B. Migraine is a recognised contraindication to combined hormonal contraceptives (stroke risk)
C. Migraine only contraindicates progestin-only methods
D. Migraine increases contraceptive failure rates via enzyme induction
E. Migraine is a contraindication only for IUDs

19. Oestrogens undergo extensive hepatic glucuronide conjugation yet maintain significant systemic bioavailability. Which pharmacokinetic process best explains this? HARD

A. Enterohepatic circulation — biliary excretion followed by intestinal deconjugation and reabsorption
B. Renal tubular reabsorption of the parent compound
C. Rapid first-pass activation in the lung
D. Protein binding to albumin prevents any hepatic uptake
E. Oestrogens bypass the liver entirely via lymphatic absorption

20. A drug is used for ER-positive breast cancer by blocking the enzyme that converts androstenedione and testosterone into oestrogens, rather than by blocking the oestrogen receptor itself. Which drug class is this? HARD

A. Aromatase inhibitors (e.g. letrozole, anastrozole)
B. Progestins
C. Antiprogestogens
D. Pure oestrogen-receptor blockers (e.g. clomiphene)
E. SERMs (e.g. tamoxifen)

A Answer key

#	Ans	Diff	Rationale
1	D	EASY	Corpus luteum secretes progesterone through the second half of the cycle after ovulation.
2	E	EASY	Ethinylestradiol is the standard synthetic oestrogen in most combined pills.
3	B	EASY	72 hours is the preferred window for levonorgestrel emergency contraception; sooner is always better.
4	B	EASY	Unopposed oestrogen with an intact uterus drives endometrial hyperplasia — hence the rule to add a progestogen.
5	B	MEDIUM	Monophasic = constant dose throughout; triphasic increases progestin over three 7-day blocks to mimic the natural cycle.
6	E	MEDIUM	Drospirenone is the 4th-generation, anti-androgenic progestin specifically used for acne/hirsutism.
7	D	HARD	Rifampin is a classic hepatic enzyme inducer that accelerates contraceptive steroid metabolism, lowering efficacy.
8	C	HARD	Raloxifene is anti-oestrogenic on both breast and endometrium (unlike tamoxifen, which is pro-oestrogenic on endometrium), so it avoids the endometrial cancer risk while still protecting bone.
9	A	MEDIUM	Progestin-only pills avoid oestrogen and do not affect milk production — the standard choice for breastfeeding, oestrogen-intolerant, or smoking patients.
10	A	MEDIUM	The subdermal etonogestrel implant provides roughly 3 years of contraception.
11	D	MEDIUM	Prior ectopic pregnancy (and pelvic inflammatory disease) are the stated contraindications to IUD placement.
12	D	MEDIUM	Progestin thickens cervical mucus, impeding sperm transport — a mechanism independent of ovulation suppression.
13	D	MEDIUM	Mifepristone + misoprostol is the standard medical regimen for pregnancy termination.
14	E	MEDIUM	Ulipristal remains effective up to 5 days (120h) post-coitus, beyond levonorgestrel's 72h window.
15	A	MEDIUM	Conjugated estrogens are a mixture of several natural estrogen hormones, as stated in the lecture.
16	E	MEDIUM	Return of fertility after stopping medroxyprogesterone acetate injections may be delayed by several months.

17	**E**	HARD	Oestrogen can stimulate growth of oestrogen-receptor-positive tumours, making a personal history of breast cancer/oestrogen-dependent neoplasm a contraindication.
18	**B**	HARD	Migraine is listed among the contraindications to combined hormonal contraceptives because of increased thromboembolic/stroke risk.
19	**A**	HARD	Oestrogens are excreted in bile, deconjugated by gut flora, and reabsorbed — enterohepatic circulation — which sustains systemic levels despite hepatic conjugation.
20	**A**	HARD	This describes an aromatase inhibitor (letrozole/anastrozole), which blocks the enzyme converting androgens to oestrogens rather than blocking the receptor.